

Research

Machine learning-based prediction of hazards fine PM2.5 concentrations: a case study of Delhi, India

Ram Pravesh Kumar¹ · Aditya Prakash¹ · Ranjit Singh¹ · Pradeep Kumar²

Received: 24 May 2024 / Accepted: 12 July 2024

Published online: 18 July 2024

© The Author(s) 2024 **OPEN**

Abstract

The air quality of many geographical locations has been deteriorating in the last decades. This deterioration in air quality has affected a very large number of people, and they have been diagnosed with asthma and other respiratory problems. Among various pollutants, PM2.5 is the major cause of numerous health-related problems. Predicting PM2.5 concentration levels using ML models of these dissolved particles might help residents and the government prepare a better prevention and safety plan that can eventually lower the risk factor. The present study is based on predicting the PM2.5 concentration of Delhi by applying various ML models based on meteorological features like wind speed, temperature, humidity, visibility, etc. For prediction of PM2.5, linear regression, decision tree regression, RF regression, KNN regression, and Lasso regression methods were employed in the present study. The model performance was assessed using various parameters, including MAE, MSE, RMSE, and R2 Score. In a comparative study of all regression models, linear regression demonstrated the most favorable outcomes. The model exhibited a superior fit to the data, evidenced by its lowest RMSE value (52.19), outperforming the random forest regression (RMSE = 94.75), K Nearest Neighbor regression (RMSE = 83.93), each of which yielded higher RMSE scores compared to linear regression. Lasso regression (RMSE = 65.20) and decision tree (RMSE = 68.22) also exhibited improved performance following linear regression. The findings of this study advocate for implementing strategies to enforce stringent emission regulations for both industrial operations and vehicular activities. Such measures are imperative for mitigating air pollution levels and subsequently curtailing its adverse impacts on public health within the region. Additionally, this study underscores the necessity for further research endeavours to explore future avenues, with the aim of garnering global attention towards addressing this pressing issue.

Keywords PM2.5 · Machine learning · Prediction · Multivariate analysis · Delhi

1 Introduction

Air, a mixture of several gases, is essential for all living organisms on Earth [1]. However, air pollution is on the rise due to extensive urbanization, industrialization, the use of automobiles, power plants, chemical activities, and some natural phenomena such as volcanic eruptions and wildfires [2–4]. Among various pollutants, particulate matter (PM) significantly contributes to air pollution [5–7]. Among other air pollutants, PM2.5 is the most hazardous constituent, with a diameter equal to or less than 2.5 μm , also less than ~ 3 per cent of human hair diameter [8]. These particles can enter the lungs deeply, irritate and erode the alveolar walls, and deteriorate lung function, making them extremely

✉ Ram Pravesh Kumar, praveshjnu@gmail.com; rpkuar@mail.jnu.ac.in | ¹School of Environmental Sciences, Jawaharlal Nehru University, New Delhi 110067, India. ²Department of Geophysics, Institute of Science, Banaras Hindu University, Varanasi 221005, India.



harmful to health [2, 9, 10]. Not only does PM_{2.5} aggravate asthma and respiratory irritation, but it can also impair lung function, increase the risk of heart disease, and cause cancer [11, 12].

Since the novel COVID also stabs the respiratory system, fine particles may increase the severity of COVID-19 contamination once they enter the lungs [13]. In the event the convergence of these toxin particles is exceptionally high in the environment, it seriously influences our well-being and may cause hazardous issues in a limited ability to focus time [8]. Various studies have established that particulate matter affects human health, including respiratory organs and human gene as well [14, 15]. The present study is about air adulteration in Delhi, the area that has been most influenced by air pollution during post-monsoon and winter seasons for the last 7–8 years [4]. Accurately estimating the concentration of small particles in the air is essential when addressing the danger of air contamination [16]. Several other factors, such as wind speed, bearing, temperature, perceivability, and so forth, add to determining the level of air pollution in the ambient environment [17].

The particulate issue is the most vicious type of air contamination uprightness of being tiny in size, they can dive deep into the lungs, circulatory systems and cerebrum, messing well-being up like cardiovascular failures, respiratory sickness, and sudden passing [18, 19]. An examination done in nine European nations uncovered that there were no protected degree of particulates and that for each increment in 10 $\mu\text{g}/\text{m}^3$ in PM₁₀, the cellular breakdown in the lungs rate increased by 22% [20]. The smallest PM_{2.5} particles are particularly hazardous to human health, increasing cellular breakdown in the lungs by approximately 36% per 10 $\mu\text{g}/\text{m}^3$ when they enter the respiratory system [20]. However, according to the World Health Organization [21], ~92% of the world's population lives in areas where the annual mean PM_{2.5} concentration exceeds 10 $\mu\text{g}/\text{m}^3$, which is the WHO's cut-off value for poor air quality [22].

Foreseeing PM_{2.5} focus is a perplexing assignment because of the unique nature, instability, and high inconstancy in the existence of pollutants and particulates in the urban environment [15, 23]. Simultaneously, having the option to show, foresee, and screen air quality is getting an ever-increasing number of applicable, particularly in metropolitan regions, because of the noticed basic effect of air contamination on residents' well-being and the climate [24]. Since the last few years, a significant percentage of the world's major cities have seen contamination levels that exceed all international standards, leading to several dangerous problems. PM_{2.5} is one of the important particulate matters that is accountable for human health concerns, even if many other reliable factors are at play [8, 10, 25, 26].

Some models have been explored to measure pollutant concentration in urban air. The statistical methods have been used (including Kalman filtering) as a single variable linear regression and time series analysis of historical meteorological data [5, 27]. However, these have not produced a good accuracy level considerably. This began a trend to use ML and deep learning (Neural network) to explore the prediction, as these can be used simultaneously for different features [5]. Model like neural networks and tree-based regression have improved accuracy significantly. Still, there are some models that give importance to earlier PM_{2.5} values; thus, when the time series factor is coupled with existing ML-based models, the accuracy level improves significantly [2, 27].

Tree-based regression strategies like Decision tree, Random Forest [28], and diverse perceptron regression strategies come into the front line for these sorts of analyses, adding a further improvement in precision level; boosting methods were additionally consolidated alongside present models [29]. For instance, XGBoost is utilized by [30] to investigate air pollution prediction using the LSTM (long momentary memory) network on Melbourne, Australia-specific air pollution data. A couple of machine learning-based models for the PM_{2.5} forecast are quantified by [31, 32]. For this, they have used Random Forest, XGBoost, and profound learning on multi-source distant detecting information to predict the PM_{2.5} levels in the metropolitan region of Tehran, Iran. It has been observed that XGBoost was the best-performing model in contrast with the other two, as far as R² Score, RMSE and MAE esteems [33].

A couple of boosting strategies, for instance, AdaBoost, is utilized frequently to improve the results of nature, delivered by different AI models [32]. There is a great deal of utilization cases for time arrangement estimation that have been helped by the boosting strategies. A gathering approach-based model abused improving in agri-business time arrangement gauging to accomplish advantageous outcomes [34]. The consolidated AdaBoost-LSTM for ocean surface temperature anticipating. An angle-supported choice tree calculation, in view of the Kalman Filter, was presented by [35]. Supported LSTMs were utilized for Internet Traffic Prediction by [36, 37]. Inclination boosting was likewise used to improve the postponement-based Reservoir Computing System exhibition [38]. AdaBoost, both the cut point decision and the traits associated with it, are anything but a tree hub. It is utilized for administered orders; however, it can likewise be reached out for relapse issues. AdaBoost represents Adaptive Boosting is a boosting calculation utilized related to the learning calculations to supplement its exhibitions, was joined with SVM for time-arrangement signal grouping in epileptic seizure findings by [39].

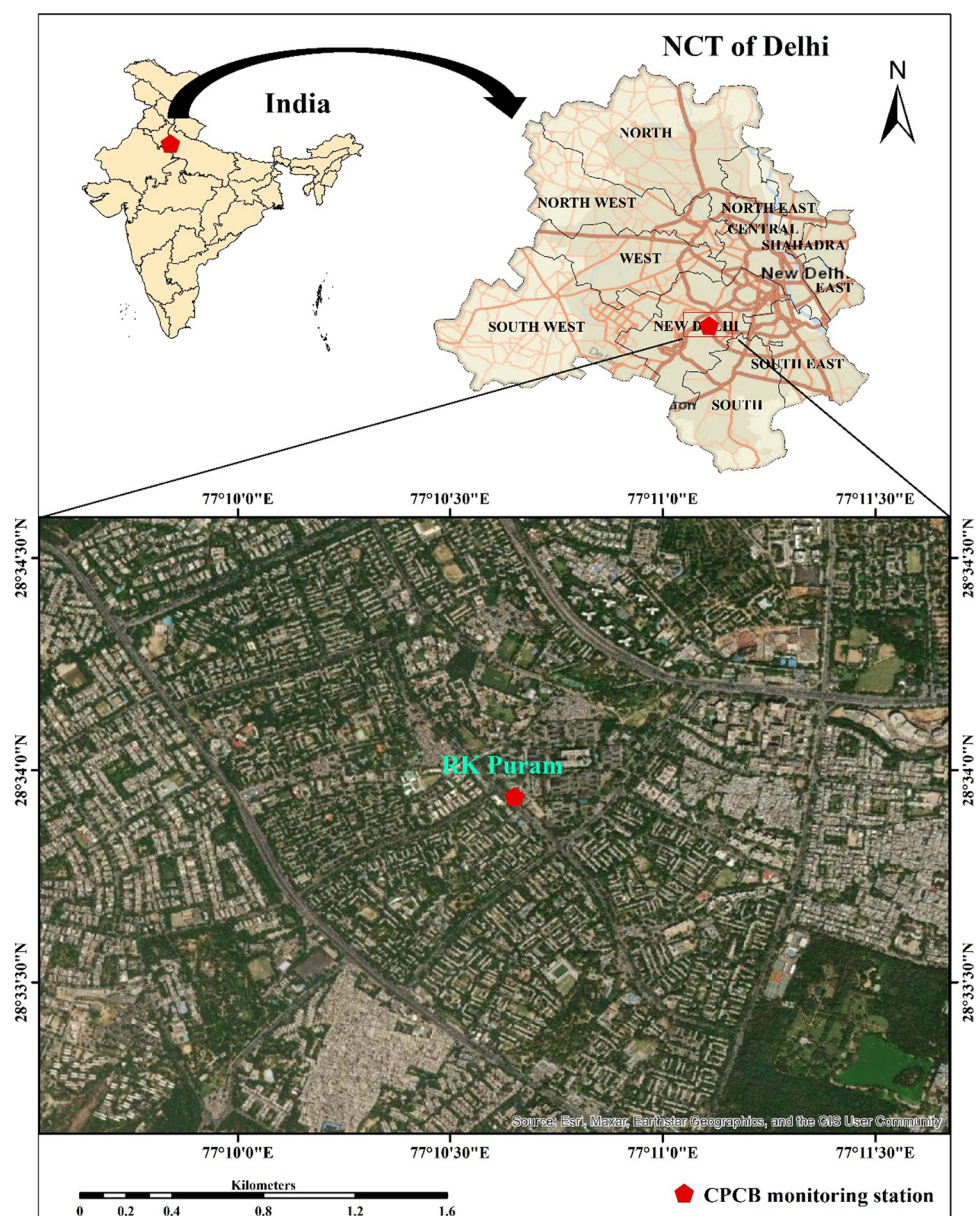
The models used for PM_{2.5} prediction, such as linear regression, random forest regression, decision tree regression, K-nearest neighbour regression, and lasso regression, differ from more advanced machine learning techniques in several ways. Linear regression is straightforward and interpretable, providing a clear relationship between the input variable quantity and the target variable, though it undertakes linearity and may not effectively capture composite outlines in the data effectively [40]. Decision tree regression is intuitive and easy to visualize, splitting the data into branches based on feature values. While useful for capturing non-linear relationships, decision trees can overfit the training data and cannot simplify well without proper pruning [41]. Using an ensemble approach that integrates many decision trees, random forest regression decreases overfitting and increases prediction accuracy. Although more robust than a single decision tree, random forests can be computationally intensive and less interpretable [42]. K-nearest neighbour regression predicts values based on the closest data points in the feature space, which is simple and effective for smaller datasets but can become slow and less accurate as the dataset grows [43]. Lasso regression is a kind of linear regression in which large coefficients are penalized as a means of regularization to prevent overfitting and enhance model generalization but may not capture complex non-linear relationships [44]. In comparison, advanced machine learning techniques offer several advantages. Ensemble methods, such as XGBoost and LightGBM, build multiple models and combine their predictions to improve accuracy. These gradient-boosting algorithms are known for their high performance and ability to handle large datasets and complex data structures more efficiently than traditional models [45]. Deep learning models, for example, recurrent neural networks (RNN) and convolutional neural networks (CNN), are accomplished automatically by recognizing intricate outlines and attributes from enormous volumes of data. CNNs are particularly effective for spatial data and image analysis, while RNNs are suitable for successive data and time series analysis. However, deep learning models need significant computational resources and proficiency to train and fine-tune [46].

In Delhi, [27] conducted a Machine Learning-based study on PM_{2.5}, revealing a significant correlation with respiratory ailments such as asthma. Their findings underscore the high prevalence of breathing-related illnesses attributed to PM_{2.5}, which dissipates into the atmosphere. Delhi has witnessed chronic respiratory health issues, especially in vulnerable people (elderly and children). Hardly any Machine learning based studies have been conducted over the Rama Krishna Puram (RK Puram) location in Delhi [27] which is home to many government officials, bureaucrats, keeping in view that more studies are needed at the given site.

Therefore, the present study aims are (a) to predict the PM_{2.5} concentration over RK Puram, Delhi, using regression and time series analysis based on both atmospheric and surface factors such as atmospheric temperature, wind speed, pressure and others (b) to find the correlation between meteorological parameters, surface factors and PM_{2.5} concentration level. This study is to be imperative for mitigating air pollution levels and curbing its adverse impacts on public health.

2 Study area

Delhi, the capital city of India became the National Capital Territory (NCT) in 1992 covers approximately 0.05% of India's total geographical area, is home to about 1.6% of the country's total population, and hosts around 8% of India's registered vehicle fleet. It is bounded by 28° 24' 16" to 28° 53' 0" N latitude and 76° 50' 20" to 77° 20' 51" E longitude, situated in the Gangetic plains of the northern part of the country, experiencing the arid and semiarid climate, with dry conditions brought on by continental air and scorching summer temperatures. Delhi experiences a variety of seasons, including warm, pleasant spring weather (February to April) with sporadic dust storms; hot, muggy summer weather (May to June) with highs of 49 °C; and a monsoon season (July to September) with sporadic dust storms, thunderstorms and powerful winds; the post-monsoon period, which spans from late September to early November and features clear skies and a cool breeze; and the chilly, foggy winter season, which spans from late November to early February and sees lows of below 10 °C. Seasonal meteorology is a major factor in aggravating pollution levels in Delhi, even though the quantity of emissions is one of the causes. Low wind conditions of less than one meter per second, cold air, and ground-based inversion are prevalent, frequently resulting in a rise in the concentration of pollutants in winter season [23]. According to the Centre for Science and Environment (CSE), Delhi has been witnessing a gradual yet consistent decline in its annual PM_{2.5} concentration since 2015–17. However, this trend reversed in 2023, with levels increasing. Moreover, Delhi recorded the highest number of smog events during the current season compared to the past six winters. In general, the winter of 2023 had the most persistently poor air quality over the last 6 years, with minimal variations in daily PM_{2.5} levels, signifying the worst air quality within this timeframe [47]. The study area map is presented in Fig. 1.

Fig. 1 Study area map

3 Materials and methodology

3.1 Data collection

Continuous ambient air quality monitoring (CAAQM) stations across the city are used to monitor and collect daily ambient air pollution data of PM_{2.5} and meteorological parameters, data was taken from (<https://app.cpcbcr.com/ccr/>) at RK Puram, spanning 2016 to 2020.

We systematically approached each stage of the present study. Initially, we clearly defined the research problem, with objectives that included understanding the relationship between meteorological parameters and PM_{2.5} concentrations. The data collection plan entails gathering both dependent (PM_{2.5} concentration) and independent (meteorological) variables. PM_{2.5} concentration, data for R.K Puram monitoring station taken from the Central Pollution Control Board (CPCB) portal. Various meteorological parameters, including V (average wind speed in Km/h), V_m (maximum sustained wind speed in Km/h), T (average temperature in °C), T_m (maximum temperature in °C), T_n (minimum temperature in °C), SLP (sea level atmospheric pressure in hPa), H (average relative humidity in %), and

VV (average visibility value in Km), are also sourced from CPCB to ensure the reliability of the datasets. The feature sets together with the corresponding units of measurement that were employed (Table 1).

Subsequently, we followed the web scraping process, which involves identifying relevant websites and systematically extracting data using tools like Beautiful Soup in Python (ML algorithms). A structured approach was adopted, iterating through each website using a for loop to retrieve monthly data. The retrieved data was then stored in an organized manner, possibly in HTML format, facilitating subsequent processing. Data processing involves defining functions to open and extract relevant information from each file, ensuring data cleanliness and consistency. Subsequently, the integration of meteorological and PM2.5 concentration data was undertaken, and datasets were aligned based on timestamps to facilitate analysis. Statistical and machine-learning techniques were then employed for data analysis. An exploratory data analysis revealed patterns and correlations, while regression analysis or other modelling techniques quantified the impact of meteorological factors on PM2.5 concentrations.

3.2 Data cleaning

The first step in making a prediction is having sufficient data for both dependent and independent variables. Then, the data will be trained on ML algorithms to predict future data. For the independent variable, we considered various meteorological variables such as wind speed (WS), temperature, relative humidity (RH) etc. and the dependent variable was PM2.5, which we predicted by various ML methods. We utilized the web scraping method for various independent variable data, and dependent variable data were obtained through CPCB websites. After procuring both data, we compiled this data in single comma-separated values file format (CSV) and used this format in ML algorithms to predict the target.

It is one of the crucial steps to be followed because the data obtained from the above process has some problems, i.e., many rows have NaN values which means empty, many have '-' values and many have 'none' as a value which was not numeric if these occur our algorithm cannot work, hence we have riden of these rows. We have removed these rows by codes in jupyter notebook. For making the dataset, we used Spyder Notebook, and for modelling and applying various algorithms, we used Jupyter Notebook and the programming language (Python). This approach ensured that the dataset utilized for analysis was free from inconsistencies, enabling robust and reliable modelling outcomes.

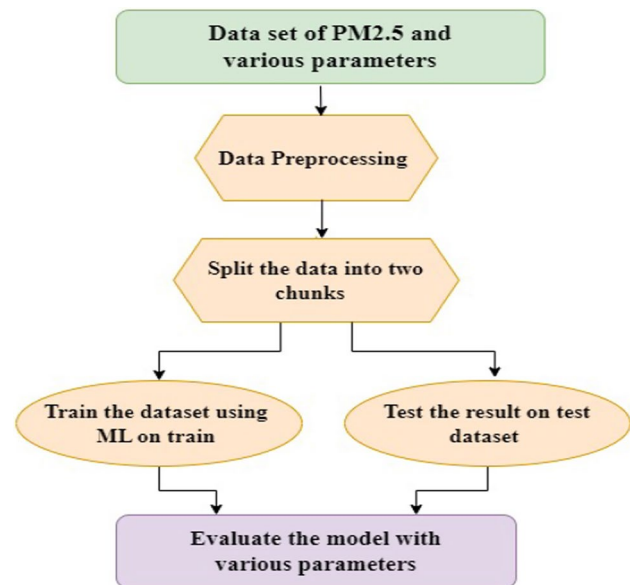
3.3 Data preprocessing

Before applying any algorithm, we made data for algorithm input, i.e., first, we converted every column value as an object datatype to a float 64 data type. Secondly, the correlation among different features must be checked. Thirdly, important information came up when modifying features/elements. Further, PM2.5 concentration levels above 600 and below 10 are considered outliers and removed [48]. Thereafter, splitting the data into two chunks, i.e., train data and test data, we applied various ML algorithms; we also tested and checked the model evolution by comparing it with the original data. The data processing flow diagram is shown in Fig. 2. We have created a data frame that includes all the features of both independent and dependent variables. Afterwards, we performed machine learning algorithms and analysis on Jupyter Notebook using Python libraries such as NumPy, Pandas, Seaborn, Matplotlib, and Scikit-learn (Sklearn).

Table 1 Parameters used for training in the model for the present study

Elements	Details	Unit
T	Average temperature	(°C)
TM	Maximum temperature	(°C)
Tm	Minimum temperature	(°C)
H	Average relative humidity	(%)
VV	Average visibility	(Km)
V	Average wind speed	(Km/h)
Vm	Maximum sustained wind speed	(Km/h)
SLP	Atmospheric pressure at sea level	(hPa)
PM2.5	Particulate Matter 2.5 μm	$\mu\text{g}/\text{m}^3$

Fig. 2 Data processing flow chart



3.4 Linear regression

Linear regression is possibly the technique on which most academicians start their ML experiences [27, 49, 50]. The core principle of machine learning involves correctly positioning one or more independent variables and the dependent variable onto a line in n -dimensional space. The n dimension usually represents the number of variables inside a dataset. Regression analysis is a technique used to identify the relationship between two or more variables. In this context, we predict the unknown variable based on its relationship with the predictor variable. The y and x variables delineate the magnitude of error associated with this relationship. This error is considered an essential parameter in finding the best relationship between two variables. In order to find the relation, we always want that line that minimizes all the sum of errors popularly named as the least square method. A certain assumption is followed by linear regression, i.e., Linear relationship, linear association between dependent variable and independent variable. It indicates that when executing autocorrelation, multicollinearity, and homoscedasticity, in a linear relationship, the change in the dependent variable remains constant for each one-unit change in the independent variable.

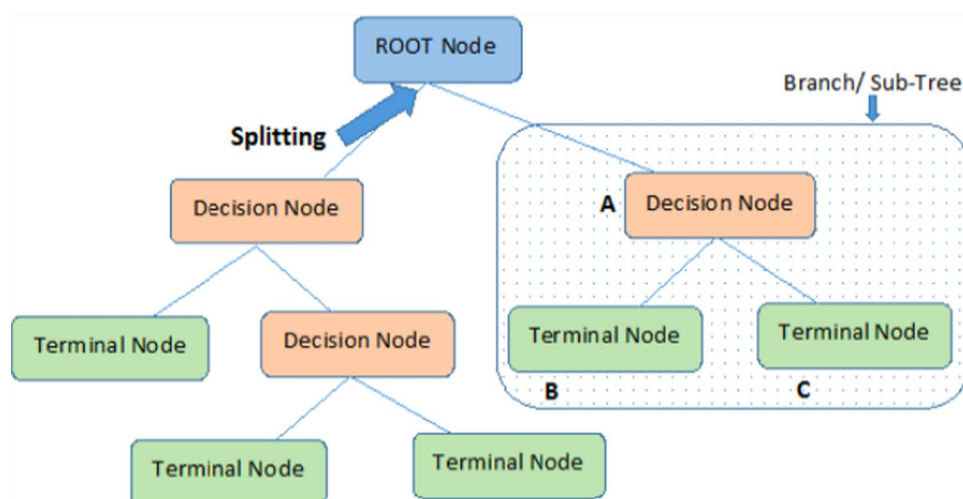
3.5 Decision tree regression

It usually works on the principle of making classification on different limit points that eventually come to be a tree; it creates various nested if-else conditions to classify the problem [51, 52]. A collection of supervised learning algorithms makes up the decision tree algorithm. Unlike other supervised learning methods, decision trees can be applied to both regression and classification problems. Therefore, using a decision tree helps create a preliminary model to learn basic decision rules from previous datasets (training data) and predict the class value of the target features.

To predict a record's class name using decision tree regression, we begin at the root of the tree. After that, we examined the negative aspects of the core feature in relation to the record's quality. We followed the branch and compared it to that worth, then leapt to the next hub based on the correlation shown in Fig. 3.

For the decision tree, various terminologies have been used, i.e., (1) The Root Node is the first node where all population is present and split according to calculation Gini or entropy or MSE. (2) Splitting, which is separating the node into further sub-nodes; (3) Decision Node- splitting when a node is divided into sub-nodes named a decision node, (4) Leaf/Terminal node- present at last as a Leaf/terminal node because further splitting is not possible or is restricted by the user, (5) Branch or Sub-Tree refers to a subset of the entire tree structure that is named as a branch or subtree and (6) Parent and Child Node: it is divided into sub-nodes, with each sub-node referred to as a parent node of its own sub-nodes. Further, in decision tree regression, the default splitting is done through MSE criteria, which is done before each splitting algorithm where MSE calculation is done. Again, it was divided with the least MSE values.

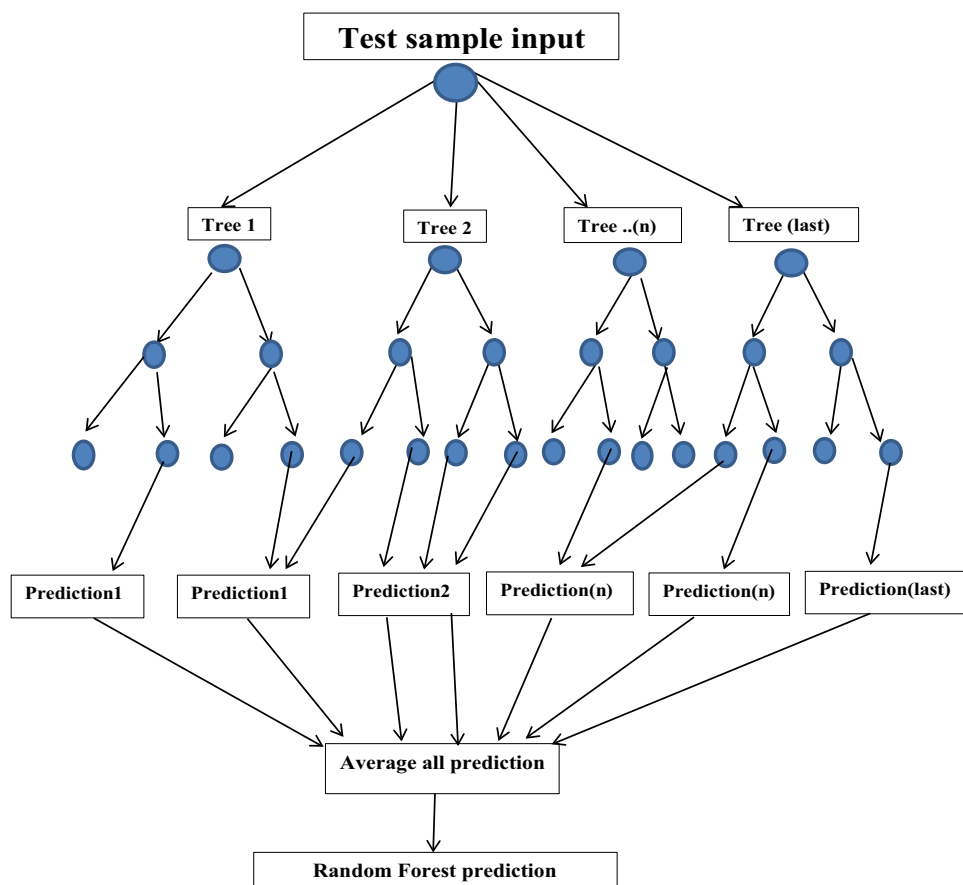
Fig. 3 General flow chart of decision tree



3.6 Random Forest (RF) regression

Regression and classification in RF are accomplished by building several decision trees during training, and the output is determined by the average prediction (regression) or the mode of classes (classification) of the cumulative trees. RF works by constructing many decision trees and observing the output class. In the case of classification, it uses the mode of classification and prediction [27, 53]. Sample tests in the random tree regression have been mentioned in Fig. 4.

Fig. 4 Random Forest prediction flow chart



The number of features can be divided at every node, and it is limited to some level as a cumulative (named as the hyperparameter). The guarantees, which have been the gathering model, are not contingent too vigorously on any individual constituents and could utilise all possibly prescient features. Each and every tree draws an arbitrary sample from the first original dataset while creating its split, but it is a further component of haphazardness that forestalls overfitting. Specifically, Trees overfit their training sets, becoming exceedingly profound and adapting to extraordinarily irregular examples. They have a low inclination but a very high change, for instance. To reduce variation, multiple deep decision trees that have each been trained on a different subset of the same training dataset are combined in an ensemble learning technique called irregular forest. This method often greatly improves the final model's overall performance but also somewhat increases bias and decreases interpretability. Woods resemble the arranging of choice tree calculation endeavors. Taking the collaboration of numerous trees along these lines improves the presentation of a solitary arbitrary tree. Although not precisely comparative, timberlands give the impacts of a K-fold cross-validation.

3.7 K Nearest Neighbor regression

KNN algorithm works based on similarity measurement. This method uses feature similarity to predict the new point, basically, it stores all the data from the training set and uses it when a new data point is introduced. It calculates the distance between the new point and each point in the training set, identifying the one with the smallest distance. In the case of prediction classification or regression, algorithms take the majority votes of the neighbor [4, 16, 27]. The distance has been calculated by the following formulas:

3.7.1 Euclidean distance

It calculates the distance between two points in Euclidean space

$$\sqrt{\sum_{i=1}^n |x_i - y_i|^2} \quad (1)$$

where x_i and y_i corresponding coordinates or values of the i^{th} dimension for two points.

3.7.2 Manhattan distance

Manhattan distance is a metric used to calculate the distance between two points in a grid-based space, like a city grid, where movement is restricted to horizontal and vertical directions, without diagonal movement.

$$\sum_{i=1}^n |x_i - y_i| \quad (2)$$

where x_i and y_i are the distance between two points in n-dimension is the sum of the distance in each dimension.

3.8 Lasso regression

In order to increase the precision and comprehensibility of the statistical model it generates, regression analysis techniques such as the Lasso (least absolute shrinkage and selection operator). We perform Lasso regression for regularization along with variable selection. This algorithm is slightly different from linear regression. The loss function (difference of predicted and actual value of data sets) of linear regression is modified in such a way as to shrink the coefficient of linear equation, is known as a regularization technique; it is used over regression methods for more precise outcomes. Lasso regression is a linear regression that employs shrinkage, in which the data values are compressed toward a mean-like central point. It penalizes less important features of your dataset by shrinking their coefficients to zero, effectively removing them. This process provides the benefits of feature selection and creates a simpler model [52, 54].

$$D = (\text{Least - square} + \text{lambda}) * \sum (\text{absolute values of the magnitude of the coefficients}). \quad (3)$$

3.9 Model evaluation

There are some criteria to see how our model is performing after predicting the values so, evaluations are done through minimum error of different kind and R2 Score tell us how close the model is fitting the curve. There are 3 types of error defined commonly each of these have its own importance and is computed with the followings:

1. Mean absolute error (MAE) [55] is the average of absolute values of error.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - y_i^{\wedge}| \quad (4)$$

where n is the number of observations/samples, y_i is the actual value of the i th observation, and $y_i^{\wedge}$ indicates the predicted value.

2. Mean squared error (MSE) i.e.mean of the squared errors.

$$MSE = \frac{1}{n} \sum_{i=1}^n |y_i - y_i^{\wedge}|^2 \quad (5)$$

where n is the number of observations/samples, y_i is the actual value of the i th observation, and $y_i^{\wedge}$ indicates the predicted value.

3. Root mean squared error (RMSE). The RMSE is compared using a third measure. It is computed using the following formulas, which show the standard deviation of the variations between expected and actual values within a dataset.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n |y_i - y_i^{\wedge}|^2} \quad (6)$$

The final performance metric utilized is the R2 Score, also known as R-squared, and sometimes referred to as the coefficient of regression. R2 Score, as described by [55]. Assesses the proximity of predicted values to actual values along a regression line. Which is calculated using the following relation:

$$R2 = 1 - \sum_{i=0}^n (y_i - y_i^{\wedge})^2 / (y_i - \mu)^2 \quad (7)$$

where the mean of the actual values is indicated by μ . The suggested model's (ET + Adaboost) performance for MAE is displayed using bar plots to show where it is. The predicted values are closer to the real ones when the MAE is lower. It is a measurement of how well the model predicts the actual values by calculating the average magnitude of the mistakes in a set of forecasts. In Table 2, the results of the proposed model are indicated in red, showing that the Mean Absolute Error (MAE) of the proposed model (ET + AdaBoost) was 43.705. This value is close to that of the DT + AdaBoost model and lower than the other models that were analyzed.

A comparative analysis presents the RMSE values of the proposed model with those of other currently used models in Table 2. It is clear that the suggested model has the lowest root mean square error (RMSE) of any of the other models, meaning that the actual values show less error than the other model's predicted values when compared to the ones predicted by the proposed model (Table 2). This emphasizes how important the suggested model is in comparison to

Table 2 Comparison of Models performed for MAE, MSE and RMSE

Model name	MAE	MSE	RMSE
1. Linear regression	48.70	3867.69	52.19
2. Decision tree regression	48.12	4654.13	68.22
3. Random Forest regression	58.43	8977.75	94.75
4. K Nearest neighbor	56.41	7044.53	83.93
5. Lasso regression	45.26	4252.16	65.20

the others. Subsequent experiments were carried out to compare the R2 Score of the proposed model with the other mentioned models. Observations from Table 2 reveal that the proposed model attains the highest R2 Score of 62.19, indicating that its prediction accuracy is substantially superior to that of other existing models on the dataset used. The summary from Table 2 collectively indicates that the proposed model is notably more effective than other existing models in forecasting the PM2.5 concentration level in polluted air.

4 Results and discussion

We have applied linear regression, decision tree regression, RF regression, KNN regression, and lasso regression algorithms as regressors to predict PM2.5 (dependent variable). The analysis revealed both linear and nonlinear relationships among various features. A colour-coded correlational matrix visually represents the degree of correlation between features. In the colour scheme utilized, red denotes the least correlation, while green indicates the highest degree of correlation. Intermediate shades between red and green signify varying degrees of correlation, with progressively greener hues indicating stronger correlations, while red signifies the least correlation. The details are shown and presented in Fig. 5.

Nevertheless, as depicted in Fig. 6, the majority of the datasets exhibit a non-normal distribution. Despite the prevailing distribution pattern, we performed a Spearman correlation analysis, which revealed statistically significant negative correlations between PM2.5 and most of the selected features: T (−0.52), TM (0.44), Tm (−0.58), H (0.12), V (−0.65), VV (−0.35), and Vm (−0.24). While a strong positive correlation is observed among all surface features. Neither PM2.5 nor surface features exhibit independent behaviour with respect to the environment. No discernible linear relationship is observed between PM2.5 and other features. The correlations between T and TM (0.96), T and Tm (0.95), TM and Tm (0.89), and VV and Vm (0.79) exhibited maximum correlation coefficients. Conversely, the correlations between V and T (0.71), V and TM (0.67), and V and Tm (0.62) demonstrated moderate correlation coefficients. Other correlations either displayed negative correlations or exhibited poor positive correlations with each other.

In light of the absence of a linear relationship between PM2.5 and other features, as illustrated in the figure, we proceeded to apply the first algorithm, linear regression. We have split data into two chunks to check how our model is performing with new data. 70% of data of (rows) was randomly trained data used on the linear regression class of the sklearn library (test_train_split). The remaining 30% of the data was used to see the accuracy of the model. After performing the models results show most of the dataset of PM2.5 concentration levels lies within 100 µg/m³ from 2016–2020 (Fig. 7).

In 2019, we found that January and February, mid-March and October, November and December have the maximum concentration of PM2.5 as compared to the rest of the month, i.e., April, May, June, July, August and September. Similar patterns were observed in the years 2016, 2017, 2018, and 2020. The distribution of PM2.5 concentrations at three-month intervals throughout the 2019 is shown (Fig. 8).

After analyzing the various graphs of PM2.5 probability density function and time series data, the density is more is between 0–100 µg/m³, and there is less density after 300 µg/m³, but we can see that density of high concentration in the November to January, and then concentration decreases slowly. In February, the density is slightly less than in January. The trend decreases up to April, but the curve is more or less very stable and constant up to October. This analysis becomes our basis for relating the PM2.5 concentrations with meteorological variables like WS, RH, temperature, visibility and atmospheric pressure.

4.1 Linear regression

This is the coefficient of each independent variable that comes after applying linear regression. This is called multiple linear regression [52, 54]. These coefficients tell us about the quantity of that independent variable that contributes to the dependent variable (PM2.5).

$$\text{PM2.5} = A1 \times T + A2 \times TM + A3 \times Tm + A4 \times \text{SLP} + A5 \times H + A6 \times VV + A7 \times Vm + c$$

Coefficient interpretation

1. Keeping all the variables fixed and a unit positive change in the T variable is associated with 3.687 times increase in PM2.5 concentrations.

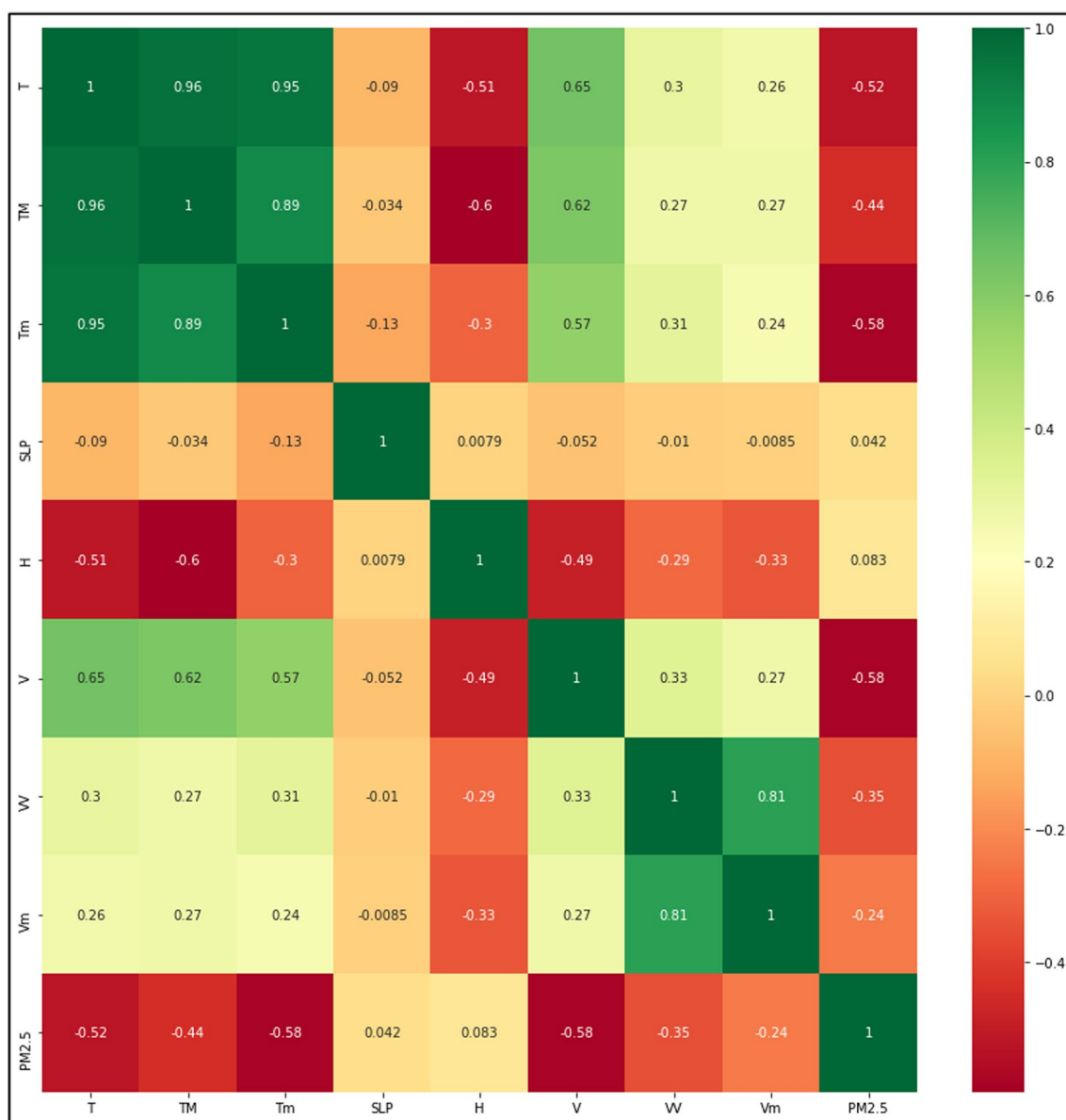


Fig. 5 Color-coded correlation metrics among PM2.5 and surface features

- Keeping all elements (Table 1) variable fixed and a unit positive change in the V variable is associated with a 48.42 decrease in PM2.5 concentrations.

We observed that the coefficient of average visibility was the highest (negatively) of the variables, and the coefficient of atmospheric pressure at sea level was the most negligible value, revealing that PM2.5 was highly dependent on average visibility and least dependent on SLP value. The R2 Score of the linear regression model was 0.519 for the train data set and 0.566 for the test dataset. We have determined the importance of the feature by using the extra tree regressor class on the sklearn package. Our results depicted that Tm has given the highest weight by extra tree regressor. While SLP has been given less importance. A similar result was observed in the linear regression model.

4.2 Decision tree regression

After applying linear regression, we have applied decision tree regression on the same data set. The very first step was dataset divided into two chunks and so that after modeling we can see the result. 70% of dataset used to train the model

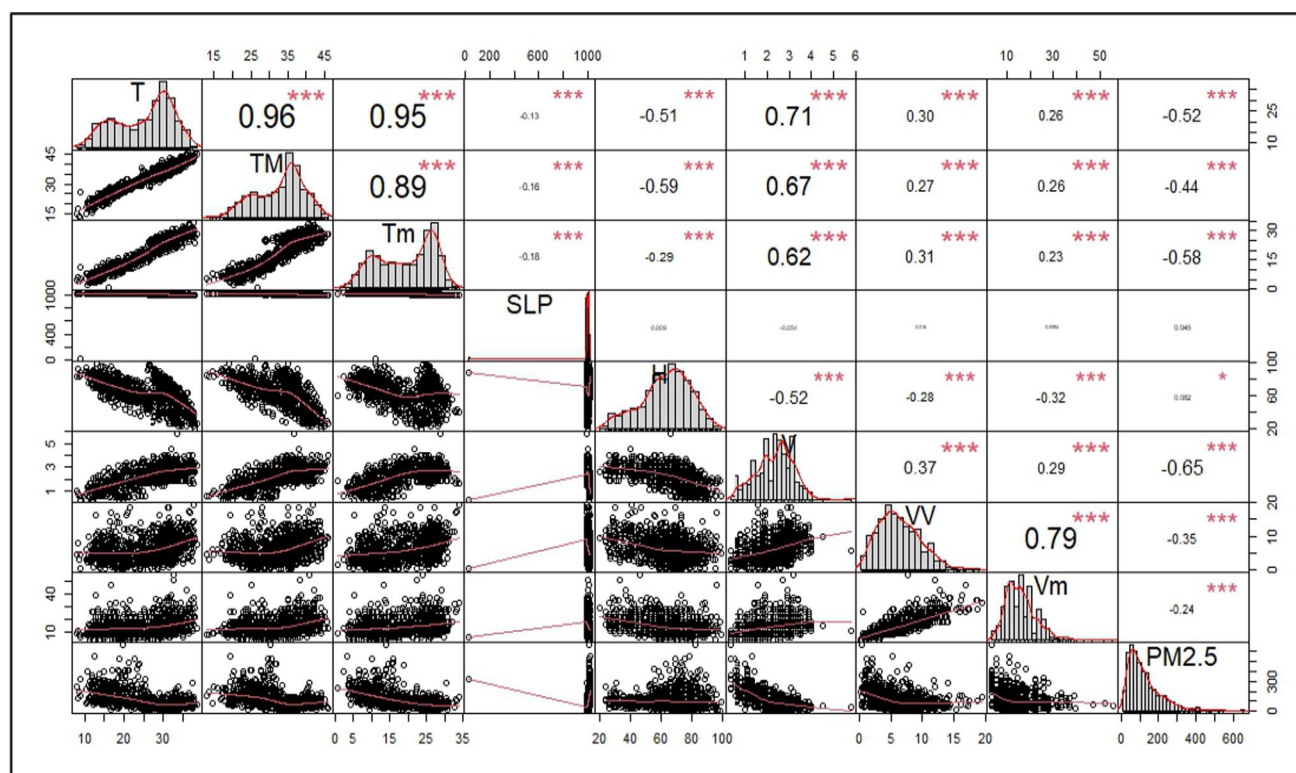


Fig. 6 pertains to the Spearman correlation chart depicting relationships between PM2.5 and T, TM, Tm, H, V, VV, SLP and Vm. ***Correlation is statistically significant at 0.05 level (2-tailed)

and 30% data used to test the model. This works on the principle of splitting the data according to minimum squared error (MSE). After splitting, it forms a tree-like structure. In the training phase, it tries to train in such a way that it can fit every data set, which is commonly known as overfitting. Because of overfitting, or the algorithm's attempt to record every data point on the training dataset, we discovered that the test dataset's R2 value was less than 1, indicating that the tests on the test dataset do not adhere to the algorithm.

4.3 Random forest regression

We have used another model, i.e., Random Forest, which works by constructing many decision trees and observing the output class in case of classification. It uses the mode of classification and means prediction in the case of regression. After performing the random forest regression, we found a very promising R2 Score (0.945) for the training data set. The R2 Score (0.620) of the test dataset. A mean Score (0.489) was observed after cross-validation.

4.4 K Nearest Neighbor regression

This algorithm gives every data point a position and compares test data to the neighbour that correspondingly predicts the value. The parameter is how algorithms measure similarity. This method uses feature similarity to predict the new point; basically, it stores all the data of the training set. When a new data point is provided, the algorithm calculates the distance between this new point and all the points in the training set. The point with the least distance to the new point is considered its nearest neighbor. R2 Score of regression on the training set is 0.6417 while on the test set it comes out to be 0.4238.

4.5 Lasso regression

Lasso regression is very similar to linear regression with little difference in cost function, which minimizes the cost function. For this method, we imported Lasso from the sklearn library. Grid search CV is used to do cross-validation. There have

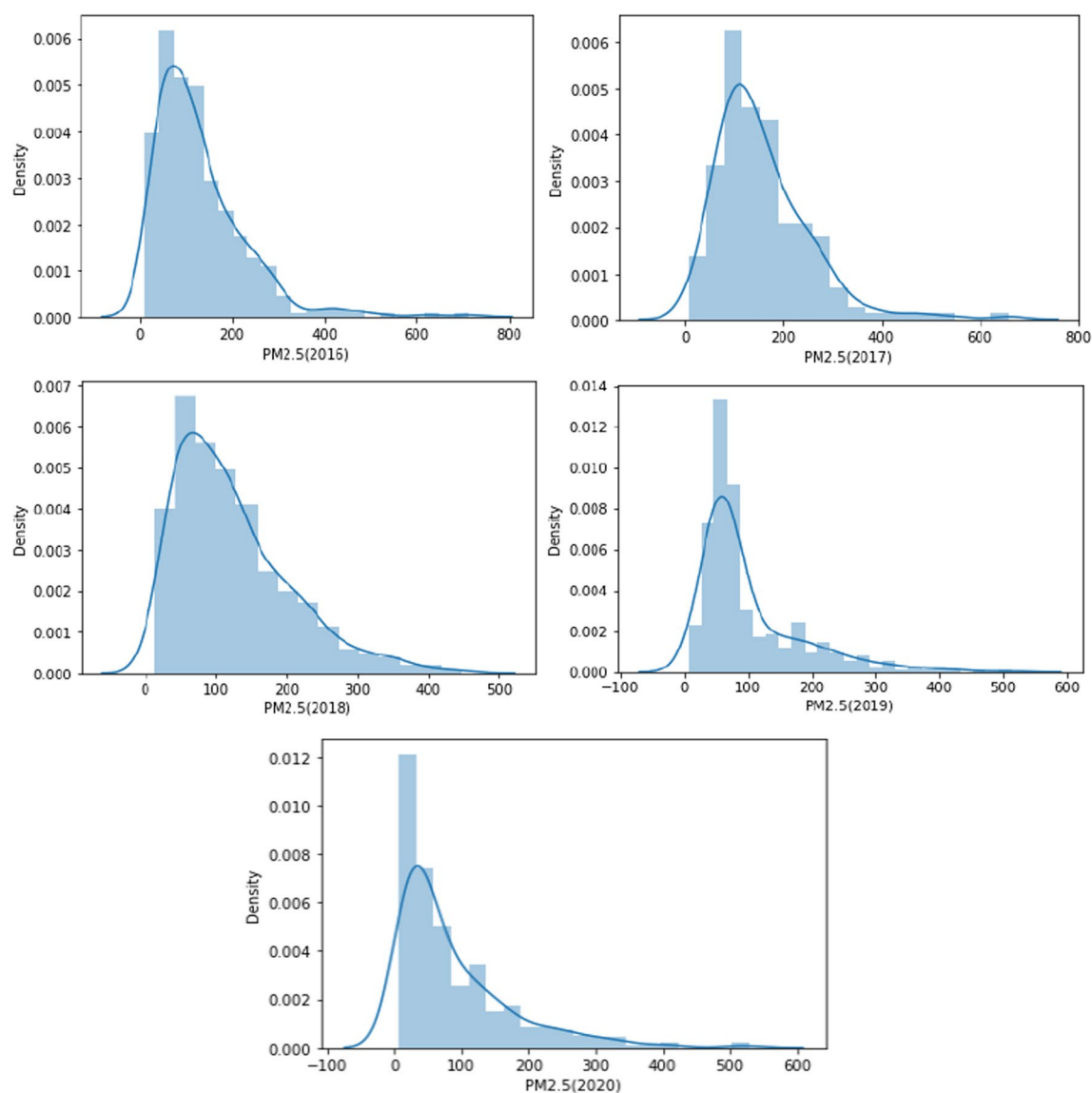
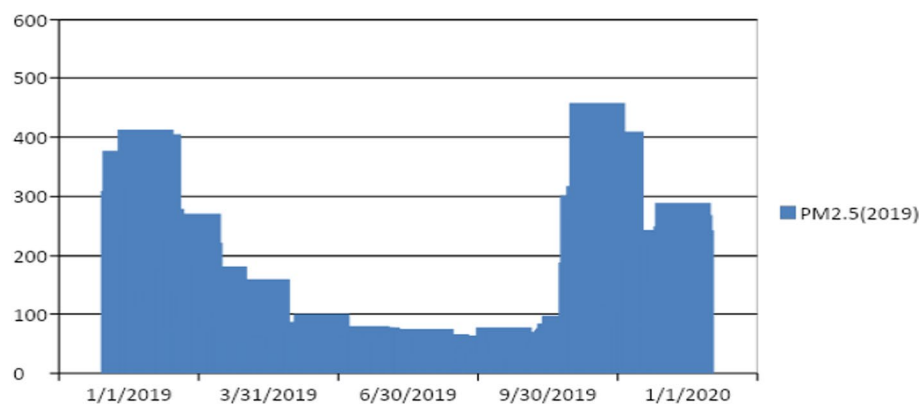


Fig. 7 Illustrates that the majority of concentrations fall within the range of 0–100 $\mu\text{g}/\text{m}^3$, with some concentrations exceeding 500 $\mu\text{g}/\text{m}^3$. The graph exhibits a gradual decrease, suggesting the presence of a distinct peak area

Fig. 8 Distribution of PM_{2.5} concentrations at 3-month intervals throughout the year 2019



been various parameters of alpha. The best alpha value we have observed 10. The scoring parameter used is a negative mean squared error (MSE). Cross-validation is a technique of resampling in different chunks and applying algorithms on each resampled data, then calculating the average score so that every data is trained and tested for more precision. The alpha value comes out to be 10, and the best score is -9960.36 .

5 Model evaluation

Figure 9 illustrates the regression analysis of Y-test values, representing actual PM2.5 concentrations (x-axis), against predicted values (y-axis). Performance metrics such as RMSE, MAE, and R2 Score are employed for evaluation. The Y-test and predicted values are roughly 45 degrees, which means that our predicted value is close to the actual value, and the density of the predicted value comes out to be a normal distributed curve. The evaluation of PM2.5 pollutant particles in Delhi Air, India, was conducted using data collected from RK Puram by CPCB from January 2016 to December 2020. During the data analysis phase, it was observed from the plot that PM2.5 concentrations are significantly elevated. They exceed $500 \mu\text{g}/\text{m}^3$ during the months of November to January, while in other months, especially during the summer season, they remain relatively low, ranging from $0\text{--}100 \mu\text{g}/\text{m}^3$ [4]. Wind influence on the data indicates that during periods of low wind speed, PM2.5 concentrations tend to be very high.

The performance analysis segment of the model shows that genuine and anticipated qualities are exceptionally near one another and structure a 45-degree slant as displayed. In this work, for the most part, three measurements have been utilized to gauge the prediction of the proposed model, which is as follows: the first is MAE, and MSE is utilized to quantify the flat partition among genuine and estimated values. RMSE is utilized as a subsequent measurement, and R2 Score is utilized as the third metric to assess the proximity of predicted values to actual values on the regression line. In the same analysis section, the proposed model is compared with other recent works, revealing that linear regression is one of the best-performing models. Figure 10 illustrates the MAE, RMSE and MSE of diverse models throughout the study (2016–2020), offering a comprehensive assessment of model performance and accuracy across the designated geographical area.

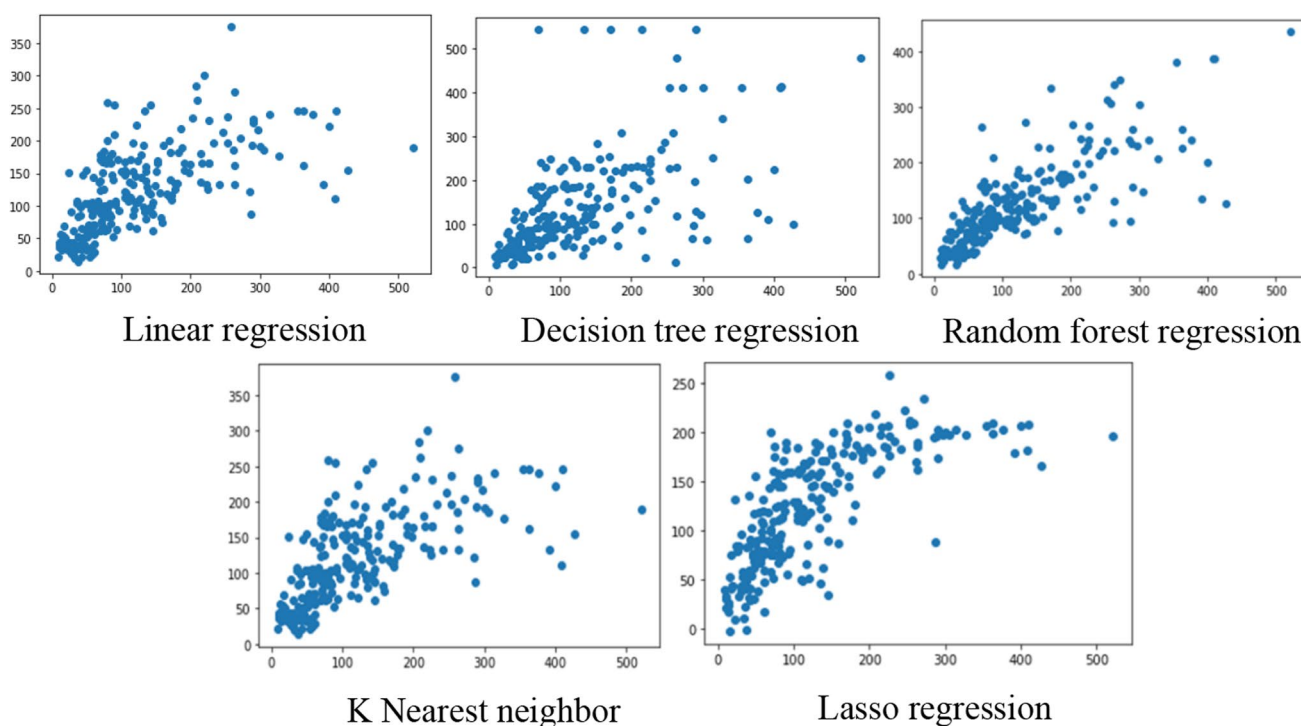
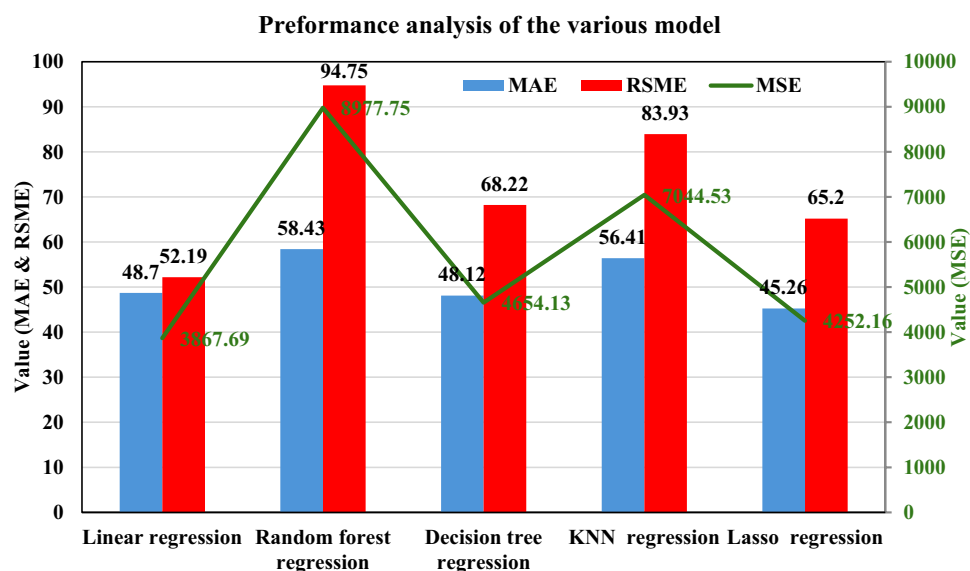


Fig. 9 Regression of Y-test value of actual value (x-axis) and predicted value (y-axis) of PM2.5 using various metrics like RMSE, MAE, and R2 Score

Fig. 10 Performance and accuracy analysis using MAE, RMSE and MSE of various models during 2016–2020



It is frequently observed that time sequence-based data can be accurately analyzed using ML-based models. In this study, we provide a successful machine learning model that predicts PM_{2.5}, a gauge of air quality in Delhi's atmosphere. After studying all the model evaluation data. In addition, linear regression has the highest R² Score and the lowest overall error (MAE, MSE, and RSME) of all five models, we may conclude that it has performed better in every parameter. The results showed that the meteorological parameter data best suited the PM_{2.5} concentration data in a linear fashion.

6 Conclusion

This study employs ML techniques to predict hazardous fine PM_{2.5} concentrations, focusing on the urban context of Delhi, India. PM_{2.5} pollution poses significant health risks, and accurate prediction models are essential for effective mitigation strategies. Leveraging historical data, meteorological parameters, and pollutant sources, various ML algorithms are evaluated for their efficacy in predicting PM_{2.5} levels. This study focuses on predicting PM_{2.5} levels in Delhi by leveraging meteorological parameters such as wind speed, temperature, humidity, and visibility by applying different models, such as linear regression, RF regression, decision tree regression, KNN regression, and lasso regression. In a comparative analysis, linear regression (RMSE = 52.19) demonstrated superior performance, displaying the lowest RMSE value and effectively fitting the data, as opposed to the random forest (RMSE = 94.75) and K nearest neighbour (RMSE = 83.93). Notably, lasso regression (RMSE = 65.20) and decision tree (RMSE = 68.22), also exhibited improved performance following linear regression. The selected models have effectively fitted the data. However, certain limitations persist. Enhanced model prediction efficacy could be achieved with larger datasets, underscoring the importance of data volume in refining predictive models. Feature selection is crucial in regression analysis, so better feature selection would improve the result. The findings contribute to advancing predictive models tailored to urban air quality management, facilitating proactive measures to combat pollution-related health hazards in densely populated regions like Delhi. Furthermore, mitigating air pollution is crucial for achieving SDG 11 (sustainable cities and communities) by enhancing public health, increasing urban resilience, and fostering sustainable urban development.

Acknowledgements Dr. RP Kumar and Mr. Ranjit Singh express their thanks to the Scientific and Engineering Research Board (SERB), Department of Science and Technology, Government of India, New Delhi, for their financial help in carrying out the present study [under the project-EEQ/2021/000699]. The authors express their sincere gratitude to the School of Environmental Sciences, Jawaharlal Nehru University, New Delhi, for providing the infrastructural facilities and conducting this research work. The authors are grateful to the Air Quality Monitoring Program of the CPCB, New Delhi, for providing the online freely available data.

Author contributions Ram Pravesh Kumar: writing—review & editing, writing—original draft, supervision, methodology, investigation, data curation, conceptualization, prepared Fig. 2. Ranjit Singh: writing—review & editing, visualization, software, prepared Figs. 1, 6, 10, methodology, data curation. Pradeep Kumar: writing—review & editing, investigation, Aditya Prakash: writing—software, formal analysis, data curation, prepared Figs. 3, 4, 5, 7, 8, 9. All authors reviewed the manuscript.

Data availability Data will be made available on request.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Zanobetti A, Franklin M, Koutrakis P, Schwartz J. Fine particulate air pollution and its components in association with cause-specific emergency admissions. *Environ Health*. 2009;8(1):58. <https://doi.org/10.1186/1476-069X-8-58>.
2. Choudhary A, Kumar P, Pradhan C, et al. Evaluating air quality and criteria pollutants prediction disparities by data mining along a stretch of urban-rural agglomeration includes coal-mine belts and thermal power plants. *Front Environ Sci*. 2023. <https://doi.org/10.3389/fenvs.2023.1132159>.
3. Pratap V, Kumar A, Tiwari S, Kumar P, Tripathi AK, Singh AK. Chemical characteristics of particulate matters and their emission sources over Varanasi during winter season. *J Atmos Chem*. 2020;77(3):83–99. <https://doi.org/10.1007/s10874-020-09405-6>.
4. Kumar RP, Singh R, Kumar P, et al. Aerosol-PM_{2.5} Dynamics: in-situ and satellite observations under the influence of regional crop residue burning in post-monsoon over Delhi-NCR, India. *Environ Res*. 2024. <https://doi.org/10.1016/j.envres.2024.119141>. (Published online 2024).
5. Kumar RP, Kashyap P, Kumar R, Pandey AK, Kumar A, Kumar K. Cancer and non-cancer health risk assessment associated with exposure to non-methane hydrocarbons among roadside vendors in Delhi, India. *Human Ecol Risk Assess*. 2020;26(5):1285–99. <https://doi.org/10.1080/10807039.2019.1570077>.
6. Kulkarni SH, Ghude SD, Jena C, et al. How much does large-scale crop residue burning affect the air quality in Delhi? *Environ Sci Technol*. 2020;54(8):4790–9. <https://doi.org/10.1021/ACS.EST.0C00329>.
7. Sinha S, Kumar R, Kumar AR, Prabhu V, Kumar RP, Tripathi AK. Elemental characteristics of respirable particulate matter in the urban atmosphere of Dehradun, Uttarakhand, India. *Curr World Environ*. 2021;16(1):319–28. <https://doi.org/10.12944/cwe.16.1.32>.
8. Pozzer A, Anenberg SC, Dey S, Haines A, Lelieveld J, Chowdhury S. Mortality attributable to ambient air pollution: a review of global estimates. *Geohealth*. 2023. <https://doi.org/10.1029/2022GH000711>.
9. Pandey AK, Kumar RP, Kumar K. Satellite and ground-based seasonal variability of NO₂ and SO₂ over New Delhi, India: In: Remote Sensing of Clouds and the Atmosphere XX SPIE. 2015;9640:212–18. <https://doi.org/10.1117/12.2193542>.
10. Kashyap P, Kumar A, Kumar RP, Kumar K. Biogenic and anthropogenic isoprene emissions in the subtropical urban atmosphere of Delhi. *Atmos Pollut Res*. 2019;10(5):1691–8. <https://doi.org/10.1016/j.apr.2019.07.004>.
11. Lewis TC, Robins TG, Dvonch JT, et al. Air pollution-associated changes in lung function among asthmatic children in Detroit. *Environ Health Perspect*. 2005;113(8):1068–75. <https://doi.org/10.1289/ehp.7533>.
12. Murray CJL, Aravkin AY, Zheng P, et al. Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet*. 2020;396(10258):1223–49. [https://doi.org/10.1016/S0140-6736\(20\)30752-2](https://doi.org/10.1016/S0140-6736(20)30752-2).
13. Santurtún A, Colom ML, Fdez-Arroyabe P, del Real Á, Fernández-Olmo I, Zarrabeitia MT. Exposure to particulate matter: direct and indirect role in the COVID-19 pandemic. *Environ Res*. 2022. <https://doi.org/10.1016/j.envres.2021.112261>.
14. Rider CF, Carlsten C. Air pollution and DNA methylation: effects of exposure in humans. *Clin Epigenet*. 2019. <https://doi.org/10.1186/s13148-019-0713-2>.
15. Kumar P, Kapur S, Choudhary A, Singh AK. Spatiotemporal variability of optical properties of aerosols over the Indo-Gangetic Plain during 2011–2015. *Indian J Phys*. 2022;96(2):329–41. <https://doi.org/10.1007/s12648-020-01987-x>.
16. Mezoue CA, Ngangmo YC, Nguia S, Lenouo A, Monkam D. Measurement of fines particle concentrations and estimation of Air Quality Index (AQI) over Northeast Douala, Cameroon. *Environ Monit Assess*. 2022. <https://doi.org/10.21203/rs.3.rs-2078709/v1>.
17. Dhaka SK, Chetna, Kumar V, et al. PM_{2.5} diminution and haze events over Delhi during the COVID-19 lockdown period: an interplay between the baseline pollution and meteorology. *Sci Rep*. 2020. <https://doi.org/10.1038/s41598-020-70179-8>.
18. US EPA. US EPA National Center for Environmental Assessment. US EPA; 2020.
19. Manisalidis I, Stavropoulou E, Stavropoulos A, Bezirtzoglou E. Environmental and health impacts of air pollution: a review. *Front Public Health*. 2020. <https://doi.org/10.3389/fpubh.2020.00014>.
20. Lee YG, Lee PH, Choi SM, An MH, Jang AS. Effects of air pollutants on airway diseases. *Int J Environ Res Public Health*. 2021. <https://doi.org/10.3390/ijerph18189905>.
21. WHO. Ambient (outdoor) air quality and health; 2022. [https://www.who.int/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)
22. Juginović A, Vuković M, Aranza I, Biloš V. Health impacts of air pollution exposure from 1990 to 2019 in 43 European countries. *Sci Rep*. 2021. <https://doi.org/10.1038/s41598-021-01802-5>.
23. Kumar R, Pandey A, Kumar R, Kashyap P, Kumar K. Assessment of source profile of nonmethane hydrocarbon in the ambient air of Metro City Delhi, India. *Curr World Environ*. 2017;12(2):326–38. <https://doi.org/10.12944/cwe.12.2.16>.

24. Ngangmo Y, Adiang CM, Choudhary A, Monkam D. Road traffic-induced particle matter dispersion in a calm wind environment at the main roundabout in Douala, central Africa. *J Air Pollut Health*. 2023. <https://doi.org/10.18502/japh.v8i1.12030>.
25. Kumar P, Pratap V, Kumar A, et al. Assessment of atmospheric aerosols over Varanasi: physical, optical and chemical properties and meteorological implications. *J Atmos Sol Terr Phys*. 2020;209: 105424. <https://doi.org/10.1016/j.jastp.2020.105424>.
26. Nahid S, Kumar RP. Spatio-temporal distribution of air pollutants and their relationship with land surface temperature over Delhi and its surroundings, India. Published online 2024. <https://doi.org/10.5194/egusphere-egu24-776>
27. Kumar S, Mishra S, Singh SK. A machine learning-based model to estimate PM_{2.5} concentration levels in Delhi's atmosphere. *Heliyon*. 2020. <https://doi.org/10.1016/j.heliyon.2020.e05618>.
28. Breiman L. Random forests. *Machine Learning*. 2001;45(1):5–32. <https://doi.org/10.1023/A:1010933404324>.
29. Kern C, Klausch T, Kreuter F. Tree-based machine learning methods for survey research. *Surv Res Methods*. 2019;13(1):73–93. <https://doi.org/10.18148/srm/2019.v13i1.7395>.
30. Sinnott R, Zhong S. Real-time route planning to reduce pedestrian pollution exposure in urban settings. In: Proceedings of the IEEE/ACM 10th international conference on big data computing, applications and technologies. BDCAT '23. Association for Computing Machinery; 2024. <https://doi.org/10.1145/3632366.3632381>
31. Joharestani MZ, Cao C, Ni X, Bashir B, Talebiefandaran S. PM_{2.5} prediction based on random forest, XGBoost, and deep learning using multisource remote sensing data. *Atmosphere (Basel)*. 2019. <https://doi.org/10.3390/atmos10070373>.
32. Lei TMT, Ng SCW, Siu SWI. Application of ANN, XGBoost, and other ML methods to forecast air quality in Macau. *Sustainability (Switzerland)*. 2023. <https://doi.org/10.3390/su15065341>.
33. Sinnott RO, Guan Z. Prediction of air pollution through machine learning approaches on the cloud. In: 2018 IEEE/ACM 5th international conference on big data computing applications and technologies (BDCAT). IEEE; 2018. p. 51–60.
34. Xiao C, Tong X, Li D, et al. Prediction of long lead monthly three-dimensional ocean temperature using time series gridded Argo data and a deep learning method. *Int J Appl Earth Obs Geoinf*. 2022. <https://doi.org/10.1016/j.jag.2022.102971>.
35. Li H, Xu F, Zhou W, et al. Development of a global gridded Argo data set with Barnes successive corrections. *J Geophys Res Oceans*. 2017;122(2):866–89. <https://doi.org/10.1002/2016JC012285>.
36. Méndez M, Merayo MG, Núñez M. Long-term traffic flow forecasting using a hybrid CNN-BiLSTM model. *Eng Appl Artif Intell*. 2023;121: 106041. <https://doi.org/10.1016/j.engappai.2023.106041>.
37. Li D. Predicting short-term traffic flow in urban based on multivariate linear regression model. *J Intell Fuzzy Syst*. 2020;39:1417–27. <https://doi.org/10.3233/JIFS-179916>.
38. Tao Z, Zhang H, Zhu C, Hao Z, Zhang X, Hu X. Design and operation of App-based intelligent landslide monitoring system: the case of Three Gorges Reservoir Region. *Geomat Nat Haz Risk*. 2019;10(1):1209–26. <https://doi.org/10.1080/19475705.2019.1568312>.
39. Al-Hadeethi H, Abdulla S, Diykh M, Deo RC, Green JH. Adaptive boost LS-SVM classification approach for time-series signal classification in epileptic seizure diagnosis applications. *Expert Syst Appl*. 2020;161: 113676. <https://doi.org/10.1016/j.eswa.2020.113676>.
40. Hastie T, Tibshirani R, Friedman J. Springer Series in Statistics The Elements of Statistical Learning Data Mining, Inference, and Prediction.
41. Breiman L, Friedman J, Stone CJ, Olshen RA. Classification and Regression Trees. Taylor & Francis; 1984. <https://books.google.co.in/books?id=JwQx-WOmSyQC>
42. Breiman L. Random forests, vol 45; 2001.
43. Altman NS. An introduction to Kernel and Nearest-Neighbor nonparametric regression. vol 46; 1992.
44. Tibshirani R. Regression shrinkage and selection via the Lasso, vol 58; 1996.
45. Chen T, Guestrin C. XGBoost: A scalable tree boosting system. In: Proceedings of the ACM SIGKDD international conference on knowledge discovery and data mining, vol 13–17. Association for Computing Machinery; 2016. p. 785–94. <https://doi.org/10.1145/2939672.2939785>.
46. Lecun Y, Bengio Y, Hinton G. Deep learning. *Nature*. 2015;521(7553):436–44. <https://doi.org/10.1038/nature14539>.
47. Times of India. <https://timesofindia.indiatimes.com/city/delhi/after-seven-years-2023-bucks-the-declining-trend-in-pm2-5-in-delhi/articleshow/106556360.cms>. Published online Jan, 2024.
48. Chowdhury S, Dey S, Di Girolamo L, Smith KR, Pillarisetti A, Lyapustin A. Tracking ambient PM_{2.5} build-up in Delhi national capital region during the dry season over 15 years using a high-resolution (1 km) satellite aerosol dataset. *Atmos Environ*. 2019;204:142–50.
49. Liang YC, Maimury Y, Chen AHL, Juarez JRC. Machine learning-based prediction of air quality. *Appl Sci (Switzerland)*. 2020;10(24):1–17. <https://doi.org/10.3390/app10249151>.
50. Czernecki B, Marosz M, Jędruszkiewicz J. Assessment of machine learning algorithms in short-term forecasting of pm₁₀ and pm_{2.5} concentrations in selected polish agglomerations. *Aerosol Air Qual Res*. 2021. <https://doi.org/10.4209/aaqr.200586>.
51. Naveen S, Upamanyu MS, Chakki K, Chandan M, Hariprasad P. Air quality prediction based on decision tree using machine learning. In: International conference on smart systems for applications in electrical sciences, ICSSSES 2023. Institute of Electrical and Electronics Engineers Inc.; 2023. <https://doi.org/10.1109/ICSSSES58299.2023.10200535>
52. Doreswamy A, Harishkumar KS, Km Y, Gad I. Forecasting air pollution particulate matter (PM_{2.5}) Using machine learning regression models. *Procedia Comput Sci*. 2020;171:2057–66. <https://doi.org/10.1016/j.procs.2020.04.221>.
53. Muljana R, Ayuningtyas LD, Daksa RP, Djamhari SF, Fiezayyan MA, Sagala NTM. Air pollution prediction using random forest classifier: a case study of DKI Jakarta. In: 2023 International Conference on Computer Science, Information Technology and Engineering (ICCoSITE); 2023. p. 428–433. <https://doi.org/10.1109/ICCoSITE57641.2023.10127759>
54. Wu T, Zhao Z, Wei H, Peng Y. Research on PM_{2.5} integrated prediction model based on lasso-RF-GAM. In: Data mining and big data: 5th international conference, DMBD 2020, Belgrade, Serbia, July 14–20, 2020, Proceedings 5. Springer; 2020. p. 83–94.
55. Botchkarev A. Evaluating performance of regression machine learning models using multiple error metrics in azure machine learning studio. Available at SSRN 3177507. Published online 2018.